```
# ANN
import numpy as np
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

data=load_iris()
x=data.data
y=data.target

x_train, x_test, y_train, y_test = train_test_split (x, y, test_size =0.2,random_state=42)

sc = StandardScaler ()
x_train = sc.fit_transform(x_train)
x_test = sc.transform(x_test)

model = Sequential ()
model.add (Dense (8, activation='relu',input_dim=4))
model.add(Dense(6,activation='relu'))
model.add(Dense(3,activation='softmax'))

model.compile(optimizer="adam",
            loss="sparse_categorical_crossentropy",
            metrics=["accuracy"])

model.fit(x_train,y_train,epochs=100)
loss,accuracy=model.evaluate(x_test,y_test)

print("Accuracy: ",accuracy)

sample=np.array([[5.1,3.5,1.4,0.2]])
sample=sc.transform(sample)
```

```
prediction=model.predict(sample)
print("Predicted class: ",np.argmax(prediction))

# CNN
import tensorflow as tf
import numpy as np
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense,Conv2D,MaxPooling2D,Flatten

(x_train,y_train),(x_test,y_test)=tf.keras.datasets.mnist.load_data(path="mnist.npz")

x_train=x_train/255.0
x_test=x_test/255.0

x_train=x_train.reshape(-1,28,28,1)
x_test=x_test.reshape(-1,28,28,1)

model=Sequential()
model.add(Conv2D(32,(3,3),activation='relu',input_shape=(28,28,1)))
model.add(MaxPooling2D(pool_size=(2,2)))
model.add(Flatten())
model.add(Dense(128,activation='relu'))
model.add(Dense(10,activation="softmax"))

model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

model.fit(x_train,y_train,epochs=2)
model.save("mnist_cnn_model.keras")

loss,accuracy=model.evaluate(x_test,y_test)
print("Accuracy: ",accuracy)
 prediction=model.predict(x_test[:1])
 print("Predicted digit: ",np.argmax(prediction))
```

Outputs:

```
(.venv) PS D:\Internship\Day52> python task.py
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.8250 - loss: 0.3768
Epoch 97/100
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.8250 - loss: 0.3750
Epoch 98/100
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.8250 - loss: 0.3731
Epoch 99/100
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.8250 - loss: 0.3713
Epoch 100/100
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.8250 - loss: 0.3698
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 311ms/step - accuracy: 0.9000 - loss: 0.3243
Accuracy:  0.8999999761581421
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 140ms/step
Predicted class:  0
```

```
Epoch 1/2
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 27s 14ms/step - accuracy: 0.9552 - loss: 0.1488
Epoch 2/2
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 50s 27ms/step - accuracy: 0.9845 - loss: 0.0504
313/313 ━━━━━━━━━━━━━━━━━━━━ 2s 6ms/step - accuracy: 0.9842 - loss: 0.0448
Accuracy:  0.9842000007629395
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 148ms/step
Predicted digit:  7
```